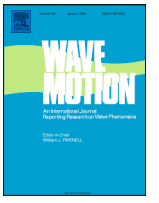
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Breathers and rogue waves on the periodic background in the Hirota and the Maxwell-Bloch equations

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ABSTRACT

For the three-component Hirota and Maxwell-Bloch equations, a general solution to the associated Lax pair is derived using the method of separation of variables and the method of variation of constants. Based on this general solution and the Darboux transformation, several types of nonlinear wave solutions are constructed. As three representative cases, breathers on the periodic background, Jacobi cn background, and Jacobi dn background are generated by taking the plane wave, the cn-periodic wave, and the dn-periodic wave as seed solutions, respectively. By choosing appropriate parameters, the corresponding dynamics and evolution of the three components are clearly revealed. For case 1, the one-breather and two-breather on the periodic background are obtained. In the case of one-breather on the periodic background, the background wave propagates along the t-axis, while the breather propagates along either the x-axis or the t-axis. For breathers on cn- and dn-periodic backgrounds, the breather travels along the t-axis, whereas the background wave propagates along the x-axis. Furthermore, in the degenerate case where the elliptic modulus m equals 1, breathers on the soliton background and two-soliton are also obtained. Next, by combining the Darboux transformation with the nonlinearization of the Lax pair, rogue waves on two types of periodic backgrounds are expressed in terms of the Jacobian elliptic functions cn and dn.

1. Introduction

The exact solution [1,2] serves as a fundamental cornerstone in the study of soliton and integrable system theory. Previous investigations have yielded remarkable insights into nonlinear waves of integrable equations, and the analysis of soliton solutions has long been a core research focus in the field of integrable systems. In recent years, by developing the Riemann-Hilbert method, researchers successfully derived the N-soliton solutions for the general n-component nonlinear Schrödinger equations and proposed a conjecture regarding the law of nonlinear wave propagation [3]. Meanwhile, the steepest descent method has been applied to investigate the long-time asymptotic behavior of solutions to the non-local LPD equation with step-like initial data, yielding profound insights into the evolutionary dynamics of such systems [4]. These advances not only enrich the theoretical framework of soliton theory but also provide important methodological references for studying more complex multi-component integrable models. Although solitons and breathers have been extensively studied against a constant background, it is widely recognized that such a scenario corresponds to

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an idealized situation. Therefore, investigating solutions featuring periodic backgrounds has garnered significant academic interest, and research on breathers over periodic backgrounds has gained considerable momentum in recent years.

In 2017, Rao and He contributed to this field through their work on the nonlocal Davey-Stewartson equation [5], building upon foundational studies by Chen and Pelinovsky [6–9]. By employing the Darboux transformation method and the nonlinearization of the Lax pair [9–17], periodic rogue wave solutions have been derived for the modified Korteweg-deVries equation [8] and the nonlinear Schrödinger equation [6,7]. Liu and Li adopted the KP hierarchy reduction method to obtain related solutions [18], while Liu, Rao, and Qiao conducted further investigations on periodic breather solutions [19]. Shi and Zhaqilao explored breather solution on a periodic background for the Kundu equation [20], and Xu and Yang carried out analogous studies for the Gross-Pitaevskii equation [21]. Huang and Guo examined breathers over different periodic backgrounds [22], and Chang and Zhaqilao investigated periodic rogue wave solutions [23] and so on.

It is well established that multi-component nonlinear equations generally exhibit rich and diverse wave solution structures. In this context, we consider the Hirota and Maxwell-Bloch equations [24]

$$u_t = \mathrm{i}\alpha(\frac{1}{2}u_{xx} + |u|^2u) + \beta(u_{xxx} + 6|u|^2u_x) + 2p, \tag{1a}$$

$$p_x = 2\mathrm{i}\omega p + 2u\eta, \tag{1b}$$

$$\eta_x = -(up^* + u^*p), \tag{1c}$$

where u, p and η are complex functions: u denotes the normalized slowly varying amplitude of the complex field envelope, p represents polarization, η stands for population inversion, and α, β and ω are real constants. Here, * indicates complex conjugation. For the Hirota and Maxwell-Bloch equations, reference [23] addressed soliton solutions, rogue waves, and the N-fold Darboux transformation. The present work aims to extend this line of research by investigating periodic breathers and breathers with periodic structure in the multi-component Hirota and Maxwell-Bloch equations.

The structure of this paper is organized as follows. In Section 2, the Lax pair and the Darboux transformation are introduced. Section 3 presents the derivation of traveling periodic solutions. In Section 4, a general solution to the Lax pair given by Eqs. (2) and (3) is constructed. Three special cases are considered by employing the plane wave, cn-periodic wave, and dn-periodic wave as seed solutions of Equation (1), leading to three corresponding general solutions. In Section 5, the one-fold Darboux transformation and the two-fold Darboux transformation are applied to obtain various types of nonlinear wave solutions for the Hirota and Maxwell-Bloch equations, including breathers on the soliton background, two-soliton, one-breather and two-breather on the periodic background, as well as breathers and breathers with periodic structure on the cn-periodic wave background, along with breathers and anti-breathers on the dn-periodic wave background. In Section 6, by combining the Darboux transformation with the nonlinearization of the Lax pair, rogue waves on the cn- and dn-periodic wave backgrounds of Jacobian elliptic functions are obtained. Section 7 summarizes the key findings and provides concluding remarks.

2. Darboux transformation

To get the Darboux transformation, we introduce the Lax pair as shown below

$$\Phi_x = U\Phi, \quad U = -\mathrm{i}\lambda\sigma_3 + U_0, \tag{2}$$

$$\Phi_t = V\Phi, \quad V = \lambda^3V_3 + \lambda^2V_2 + \lambda V_1 + V_0 + \mathrm{i}\frac{1}{\lambda+\omega}V_{-1}, \tag{3}$$

where

$$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad U_0 = \begin{pmatrix} 0 & u \\ -u^* & 0 \end{pmatrix}, \quad V_{-1} = \begin{pmatrix} \eta & -p \\ -p^* & -\eta \end{pmatrix},$$

$$V_0 = \begin{pmatrix} \frac{\alpha}{2}\mathrm{i}|u|^2 - \beta(uu_t^* - u_tu^*) & 2\beta|u|^2u + \frac{\alpha}{2}\mathrm{i}u_t + \beta u_{tt} \\ -2\beta|u|^2u + \frac{\alpha}{2}\mathrm{i}u_t^* - \beta u_{tt}^* & -\frac{\alpha}{2}\mathrm{i}|u|^2 + \beta(uu_t^* - u_tu^*) \end{pmatrix},$$

$$V_1 = \begin{pmatrix} -2\beta\mathrm{i}|u|^2 & \alpha u - 2\beta\mathrm{i}u_t \\ -\alpha u^* - 2\beta\mathrm{i}u_t^* & 2\beta\mathrm{i}|u|^2 \end{pmatrix}, \quad V_2 = \begin{pmatrix} -\alpha\mathrm{i} & -4\beta u \\ 4\beta u^* & \alpha\mathrm{i} \end{pmatrix}, \quad V_3 = \begin{pmatrix} 4\mathrm{i}\beta & 0 \\ 0 & -4\mathrm{i}\beta \end{pmatrix},$$

where λ stands for a parameter and $\Phi = (\phi_1, \phi_2)^{\mathrm{T}}$ is an eigenfunction vector. Equation (1) can be obtained through the compatibility condition $U_t - V_x + [U, V] = 0$. Next, we construct the Darboux transformation.

According to the Lax pair Eq. (2) and (3), we lead to the following gauge transformation

$$\Phi' = R\Phi = (\lambda I - S), \tag{4}$$

where

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad S = \begin{pmatrix} s_{11} & s_{12} \\ s_{21} & s_{22} \end{pmatrix},$$

and $\Phi^{'}$ must be satisfied

$$\Phi_x^{'} = U^{'}\Phi^{'}, \quad R_x + RU = U^{'}R, \tag{5}$$

$$\Phi_t^{'} = V^{'}\Phi^{'}, \quad R_t + RV = V^{'}R. \tag{6}$$

By calculating Eq. (5) and collecting the coefficients of λ, the following equation is exported

$$u^{'} = u - 2\mathrm{i}s_{12}, \tag{7}$$

$$s_{21} = -s_{12}^{*}, \tag{8}$$

$$S_t = \begin{pmatrix} 0 & u \\ -u^{*} & 0 \end{pmatrix} S - S \begin{pmatrix} 0 & u \\ -u^{*} & 0 \end{pmatrix} + \mathrm{i}[S, \sigma_3]S. \tag{9}$$

We suppose

$$S = H\Lambda H^{-1}, \tag{10}$$

where

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \quad H = \begin{pmatrix} \Phi_{1,1} & \Phi_{1,2} \\ \Phi_{2,1} & \Phi_{2,2} \end{pmatrix}.$$

To make S satisfy Eq. (8) and (9), we can especially define [24]

$$H = \begin{pmatrix} \Phi_{1,1} & \Phi_{1,2} \\ -\Phi_{1,2}^{*} & \Phi_{1,1}^{*} \end{pmatrix}. \tag{11}$$

According to Ref[25]., the one-fold Darboux transformation and the two-fold Darboux transformation can be represented as

$$u[1] = u - 2\mathrm{i}\frac{(\lambda_1 - \lambda_1^{*})\Phi_{1,1}\Phi_{2,1}^{*}}{|\Phi_{1,1}|^2 + |\Phi_{2,1}|^2}, \quad u[2] = u - 2\mathrm{i}\frac{\begin{vmatrix} \Phi_{1,1} & \Phi_{2,1} & \lambda_1\Phi_{1,1} & -\lambda_1^2\Phi_{1,1} \\ \Phi_{1,2} & \Phi_{2,2} & \lambda_2\Phi_{1,2} & -\lambda_2^2\Phi_{1,2} \\ -\Phi_{2,1}^{*} & \Phi_{1,1}^{*} & -\lambda_1^{*}\Phi_{2,1}^{*} & -\lambda_1^{*2}\Phi_{2,1}^{*} \\ -\Phi_{2,2}^{*} & \Phi_{1,2}^{*} & -\lambda_2^{*}\Phi_{2,2}^{*} & -\lambda_2^{*2}\Phi_{2,2}^{*} \end{vmatrix}}{\begin{vmatrix} \Phi_{1,1} & \Phi_{2,1} & \lambda_1\Phi_{1,1} & \lambda_1\Phi_{2,1} \\ \Phi_{1,2} & \Phi_{2,2} & \lambda_2\Phi_{1,2} & \lambda_2\Phi_{2,2} \\ -\Phi_{2,1}^{*} & \Phi_{1,1}^{*} & -\lambda_1^{*}\Phi_{2,1}^{*} & \lambda_1^{*}\Phi_{1,1}^{*} \\ -\Phi_{2,2}^{*} & \Phi_{1,2}^{*} & -\lambda_2^{*}\Phi_{2,2}^{*} & \lambda_2^{*}\Phi_{1,2}^{*} \end{vmatrix}}. \tag{12}$$

3. Traveling periodic solutions of the equations

First, we consider the following form

$$u(x,t) = Q(x)e^{\mathrm{i}\kappa t}, \tag{13}$$

where κ is denoted as a real constant and $Q(x)$ is denoted as the real period function. These quantities satisfy the equation

$$Q_{xx} + 2Q^3 - a_0Q = 0, \quad Q_x^2 = -Q^4 + a_0Q^2 + b_0, \tag{14}$$

with a_0 and b_0 being two real constants.
By calculating Eq. (13) and (1a), the following equation can be obtained

$$-2p + \frac{1}{2}\mathrm{i}e^{\mathrm{i}\kappa t}(2cQ - 2\alpha Q^3 + 12\mathrm{i}\beta Q^2Q_x - \alpha Q_{xx} + 2\mathrm{i}\beta Q_{xxx}) = 0. \tag{15}$$

Making a simple calculation of Eq. (15), we get

$$p(x,t) = \frac{1}{4}\mathrm{i}e^{\mathrm{i}\kappa t}(2\kappa - \alpha a_0)Q - \frac{1}{2}e^{\mathrm{i}\kappa t}(\beta a_0)Q_x. \tag{16}$$

Substituting Eqs. (13) and (16) into (1c), we obtain

$$\eta(x,t) = \frac{1}{2}a_0\beta Q^2 + d(t), \tag{17}$$

where $d(t)$ represents a constant that is only related to t. Further substitution of Eqs. (13), (16) and (17) into (1b) leads to

$$\kappa = \frac{1}{2}a_0(\alpha - 4\beta\omega), \tag{18}$$

$$d(t) = -\frac{1}{4}a_0\beta(a_0 + 5\omega^2). \tag{19}$$

By calculation, the solutions of Equation (1) are shown below

$$u_0 = q_0 e^{\mathrm{i}\kappa_0 t}, \quad \kappa_0 = \frac{a_0(-a_0\beta + 2q_0^2\beta + \omega(\alpha - 4\beta\omega))}{2\omega}, \tag{20}$$

$$u_{\mathrm{cn}} = m\mathrm{cn}(x;m)e^{\mathrm{i}\kappa_1 t}, \quad \kappa_1 = -\frac{1}{2}\alpha + m^2\alpha + 2\beta\omega - 4m^2\beta\omega, \tag{21}$$

$$u_{\mathrm{dn}} = \mathrm{dn}(x;m)e^{\mathrm{i}\kappa_2 t}, \quad \kappa_2 = \alpha - \frac{1}{2}m^2\alpha - 4\beta\omega + 2m^2\beta\omega, \tag{22}$$

where $m \in (0,1)$ is represented as the elliptic modulus. q_0 is a constant. For Eq. (21), $a_0 = 2m^2 - 1; b_0 = m^2 - m^4$. For Eq. (22), $a_0 = 2 - m^2; b_0 = m^2 - 1$. cn and dn are the Jacobian elliptic functions and Eqs. (21) and (22) satisfy the elliptic Eq. (14).

4. A general solution

First, to facilitate subsequent calculations, we introduce the following gauge transformation

$$\Psi = E\Phi, \tag{23}$$

in which,

$$\Psi = \begin{pmatrix}\Psi_1\\ \Psi_2\end{pmatrix}, \quad E = \begin{pmatrix} e^{-\frac{1}{2}\mathrm{i}\kappa t} & 0\\ 0 & e^{\frac{1}{2}\mathrm{i}\kappa t}\end{pmatrix}, \tag{24}$$

the Lax pair can be transformed to

$$\Psi_x = U_1\Psi, \quad E_x + EU = U_1 E, \tag{25}$$

$$\Psi_t = V_1\Psi, \quad E_t + EV = V_1 E. \tag{26}$$

For convenience, we set $\lambda = \mathrm{i}\rho$. Through straightforward calculation, we obtain

$$U_1 = \begin{pmatrix}\rho & Q(x)\\ -Q(x) & -\rho\end{pmatrix}, \tag{27}$$

$$V_1 = \begin{pmatrix} \frac{1}{4}(\tilde{a} + (2\mathrm{i}\alpha + 8\beta\rho + \frac{2a_0\beta}{\rho - \mathrm{i}\omega})Q(x)^2 + \tilde{s}) & -gQ(x) + 2\beta Q(x)^3 + \frac{1}{2}wQ(x)_x + \beta Q(x)_{xx}\\ gQ(x) - 2\beta Q(x)^3 + \frac{1}{2}wQ(x)_x - \beta Q(x)_{xx} & \frac{1}{4}(-\tilde{a} - (2\mathrm{i}\alpha + 8\beta\rho + \frac{2a_0\beta}{\rho - \mathrm{i}\omega})Q(x)^2 - \tilde{s})\end{pmatrix}, \tag{28}$$

where

$$g = \frac{\mathrm{i}(2\kappa - a_0\alpha + 4\rho(\mathrm{i}\alpha + 4\beta\rho)(\mathrm{i}\rho + \omega))}{4(\rho - \mathrm{i}\omega)}, \quad w = \mathrm{i}\alpha + 4\beta\rho + \frac{a_0\beta}{\rho - \mathrm{i}\omega},$$

$$\tilde{a} = -2\mathrm{i}\kappa + 4\mathrm{i}\alpha\rho^2 + 16\beta\rho^3, \quad \tilde{s} = -\frac{a_0\beta(a_0 + 4\omega^2)}{\rho - \mathrm{i}\omega}.$$

Following Ref[26]., a general solution $(\Psi_1, \Psi_2)^{\mathrm{T}}$ of the Lax pair Eq. (2) and (3) was derived through the method of separation of variables and the method of constant transformation

$$\begin{pmatrix}\Psi_1\\ \Psi_2\end{pmatrix} = \Upsilon_1(x)\begin{pmatrix}G_1(x)e^{-\mathrm{i}\varsigma t}\\ e^{-\mathrm{i}\varsigma t}\end{pmatrix} + \Upsilon_2(x)\begin{pmatrix}e^{\mathrm{i}\varsigma t}\\ G_2(x)e^{\mathrm{i}\varsigma t}\end{pmatrix}, \tag{29}$$

where

$$G_1(x) = \frac{4(gQ(x) - 2\beta Q(x)^3 - \frac{1}{2}wQ(x)_x - \beta Q(x)_{xx})}{4\mathrm{i}\varsigma + o},$$

$$G_2(x) = \frac{4(gQ(x) - 2\beta Q(x)^3 + \frac{1}{2}wQ(x)_x - \beta Q(x)_{xx})}{4\mathrm{i}\varsigma + o},$$

$$\varsigma = \mathrm{i}\tau\sqrt{(a_0 - 4\rho^2)^2 - 4(a_0Q(x)^2 + Q(x)^4 + Q(x)_x^2)},$$

$$\tau = \frac{k'}{4(\rho - \mathrm{i}\omega)}, \quad k' = a_0\beta + (\mathrm{i}\alpha + 4\beta\rho)(\rho - \mathrm{i}\omega),$$

$$o = \tilde{a} + \tilde{s} + (2\mathrm{i}\alpha + 8\beta\rho + \frac{2a_0\beta}{\rho - \mathrm{i}\omega})Q(x)^2,$$

$$\Upsilon_1(x) = c_1((\rho - \mathrm{i}\omega)(\tilde{a} + \tilde{s} + 4\mathrm{i}\varsigma + 2k'))^{\frac{w(\rho - \mathrm{i}\omega)}{k'}} e^{-\rho x - 4\int \frac{Q(x)^2(g - a_0\beta)}{\tilde{a} + \tilde{s} + 4\mathrm{i}\varsigma + (2\mathrm{i}\alpha + 8\beta\rho + \frac{2a_0\beta}{\rho - \mathrm{i}\omega})Q(x)^2}\mathrm{d}x},$$

$$\Upsilon_2(x) = c_2((\rho - \mathrm{i}\omega)(\tilde{a} + \tilde{s} + 4\mathrm{i}\varsigma + 2k'))^{\frac{w(\rho-\mathrm{i}\omega)}{k'}} e^{\rho x + 4\int \frac{Q(x)^2(g - a_0\beta)}{\tilde{a}+\tilde{s}+4\mathrm{i}\varsigma+(2\mathrm{i}\alpha+8\beta\rho+\frac{2a_0\beta}{\rho-\mathrm{i}\omega})Q(x)^2}\mathrm{d}x},$$

and c_1, c_2 are arbitrary constants. $\Upsilon_1(x), \Upsilon_2(x)$ satisfy the following equations

$$\Upsilon_1'(x) = \Upsilon_1(x)(-\rho - Q(x)G_1(x)), \tag{30}$$

$$\Upsilon_2'(x) = \Upsilon_2(x)(\rho + Q(x)G_2(x)). \tag{31}$$

Through the following three special cases, nonlinear wave solutions of Equation (1) are derived.

Case 1 we take $u_0 = q_0 e^{\mathrm{i}\kappa_0 t}$ as the seed solution, $Q = q_0, \kappa_0 = \frac{a_0(-a_0\beta+2q_0^2\beta+\omega(\alpha-4\beta\omega))}{2\omega}$ and

$$U_{10} = \begin{pmatrix} \rho & q_0 \\ -q_0 & -\rho \end{pmatrix}, \tag{32}$$

$$V_{10} = \begin{pmatrix} (g')^{-1}(\mathrm{i}a_0^2\beta\rho + 2w' - \tilde{a}') & (g')^{-1}\mathrm{i}q_0(a_0^2\beta + \tilde{s}' - 2\mathrm{i}a_0\beta(q_0^2 - 2\omega^2)) \\ (g')^{-1}\mathrm{i}q_0(-a_0^2\beta + \tilde{a}' + 2\mathrm{i}a_0\beta(q_0^2 - 2\omega^2)) & -(g')^{-1}(\mathrm{i}a0^2\beta\rho + 2w' - \tilde{a}') \end{pmatrix}, \tag{33}$$

where

$g' = 4\omega(\rho - \mathrm{i}\omega), \quad w' = (\alpha - 4\mathrm{i}\beta\rho)(q_0^2 + 2\rho^2)(\mathrm{i}\rho\omega + \omega^2),$

$\tilde{a}' = \mathrm{i}a_0(2q_0^2\beta\rho + \omega(\alpha\rho - \mathrm{i}\alpha\omega - 4\beta\rho\omega)), \quad \tilde{s}' = 4\omega(2q_0^2\beta + \rho(\mathrm{i}\alpha + 4\beta\rho))(\mathrm{i}\rho + \omega),$

then we can get

$$\begin{pmatrix} \Psi_{10} \\ \Psi_{20} \end{pmatrix} = \begin{pmatrix} \Upsilon_{10}(x)G_{10}(x)e^{-\mathrm{i}\varsigma_0 t} + \Upsilon_{20}(x)e^{\mathrm{i}\varsigma_0 t} \\ \Upsilon_{10}(x)e^{-\mathrm{i}\varsigma_0 t} + \Upsilon_{20}(x)G_{20}(x)e^{\mathrm{i}\varsigma_0 t} \end{pmatrix}, \tag{34}$$

where

$G_{10}(x) = G_{20}(x) = \frac{q_0(-a_0^2\beta + y' + 2a_0\beta(q_0^2 - 2\omega^2))}{z+(\mathrm{i}\alpha+4\beta\rho)(-2q_0^2-4\rho^2+r)(\mathrm{i}\rho\omega+\omega^2+p)},$

$\Upsilon_{10}(x) = c_1\sqrt{\frac{1}{\omega}(f - \mathrm{i}a_0 j)}e^{-x(\rho + \frac{q_0^2(-a_0^2\beta+4\rho(\mathrm{i}\alpha+4\beta\rho)(\mathrm{i}\rho\omega+\omega^2)+2a_0\beta(q_0^2+2\mathrm{i}\rho\omega))}{z+(\mathrm{i}\alpha+4\beta\rho)(-2q_0^2-4\rho^2+r)(\mathrm{i}\rho\omega+\omega^2)+p})},$

$\Upsilon_{20}(x) = c_2\sqrt{\frac{1}{\omega}(f - \mathrm{i}a_0 j)}e^{x(\rho + \frac{q_0^2(-a_0^2\beta+4\rho(\mathrm{i}\alpha+4\beta\rho)(\mathrm{i}\rho\omega+\omega^2)+2a_0\beta(q_0^2+2\mathrm{i}\rho\omega))}{z+(\mathrm{i}\alpha+4\beta\rho)(-2q_0^2-4\rho^2+r)(\mathrm{i}\rho\omega+\omega^2)+p})},$

$r = \sqrt{a_0^2 - 4a_0(q_0^2 + 2\rho^2) + 4(q_0^4 + 4\rho^4)}, \quad z = a_0^2\beta\rho - 2a_0q_0^2\beta\rho, \quad p = a_0\omega(-\alpha\rho + \mathrm{i}\beta r + \mathrm{i}\alpha\omega + 4\beta\rho\omega),$

$f = \mathrm{i}a_0^2\beta\rho + (\alpha - 4\mathrm{i}\beta\rho)(2q_0^2 + 4\rho^2 - r)(\mathrm{i}\rho\omega + \omega^2), \quad j = 2q_0^2\beta\rho + \omega(\alpha\rho - \mathrm{i}\beta r - \mathrm{i}\alpha\omega - 4\beta\rho\omega).$

Case 2 we take $u_{\mathrm{cn}} = m\mathrm{cn}(x;m)e^{\mathrm{i}\kappa_1 t}$ as the seed solution, $Q = m\mathrm{cn}(x;m), \kappa_1 = -\frac{1}{2}\alpha + m^2\alpha + 2\beta\omega - 4m^2\beta\omega$ and

$$U_{1cn} = \begin{pmatrix} \rho & m\mathrm{cn}(x;m) \\ -m\mathrm{cn}(x;m) & -\rho \end{pmatrix}, \tag{35}$$

$$V_{1cn} = \begin{pmatrix} \tau(1 - 2m^2 + 4\rho^2 + 2m^2\mathrm{cn}^2(x;m)) & 4m\rho\tau\mathrm{cn}(x;m) - 2m\tau\mathrm{dn}(x;m)\mathrm{sn}(x;m) + \iota \\ -4m\rho\tau\mathrm{cn}(x;m) - 2m\tau\mathrm{dn}(x;m)\mathrm{sn}(x;m) - \iota & -\tau(1 - 2m^2 + 4\rho^2 + 2m^2\mathrm{cn}^2(x;m)) \end{pmatrix}, \tag{36}$$

where

$$\tau = \frac{\alpha(\mathrm{i}\rho + \omega) + \beta(-1 + 2m^2 + 4\rho^2 - 4\mathrm{i}\rho\omega)}{4(\rho - \mathrm{i}\omega)}, \quad \iota = (1 - m)m^2\beta\mathrm{cn}(x;m)\mathrm{sn}^2(x;m),$$

then we can get

$$\begin{pmatrix} \Psi_{1cn} \\ \Psi_{2cn} \end{pmatrix} = \begin{pmatrix} \frac{-m(2\tau\mathrm{dn}(x;m)\mathrm{sn}(x;m)+\mathrm{cn}(x;m)\sigma)}{\tau(2m^2\mathrm{sn}^2(x;m)-\delta)}\Upsilon_{1cn}(x)e^{-\frac{\mathrm{i}}{2}\tau\chi t} + \Upsilon_{2cn}(x)e^{\frac{\mathrm{i}}{2}\tau\chi t} \\ \Upsilon_{1cn}(x)e^{-\frac{\mathrm{i}}{2}\tau\chi t} + \frac{-m(-2\tau\mathrm{dn}(x;m)\mathrm{sn}(x;m)+\mathrm{cn}(x;m)\sigma)}{\tau(2m^2\mathrm{sn}^2(x;m)-\delta)}\Upsilon_{2cn}(x)e^{\frac{\mathrm{i}}{2}\tau\chi t} \end{pmatrix}, \tag{37}$$

where

$\sigma = -4\rho\tau + (m-1)m\beta\mathrm{sn}^2(x;m), \quad \delta = 1 + 4\rho^2 + \frac{\mathrm{i}}{2}\chi, \quad \chi = 2\sqrt{8(2m^2-1)\rho^2 - 16\rho^4 - 1},$

$\Upsilon_{1cn}(x) = c_1(2m^2\mathrm{sn}^2(x;m) - \delta)^{\frac{1}{2}}e^{\rho(x+(4m^2\delta^{-1}-2)\Pi)+N(x)},$

$\Upsilon_{1cn}(x) = c_2(2m^2\mathrm{sn}^2(x;m) - \delta)^{\frac{1}{2}}e^{-\rho(x+(4m^2\delta^{-1}-2)\Pi)+N(x)},$

$\Pi(s,\rho,m) = \int_0^s \frac{1}{(1-\rho\xi^2)\sqrt{1-\xi^2}\sqrt{1-\rho^2\xi^2}}\mathrm{d}\xi, \quad N(x) = \int \frac{(m-1)m^3\beta\mathrm{cn}^2(x;m)\mathrm{sn}^2(x;m)}{\tau(2m^2\mathrm{sn}^2(x;m)-\delta)}\mathrm{d}x.$

Case 3 we take $u_{\mathrm{dn}} = \mathrm{dn}(x;m)e^{\mathrm{i}\kappa_2 t}$ as the seed solution, $Q = \mathrm{dn}(x;m), \kappa_2 = \alpha - \frac{1}{2}m^2\alpha - 4\beta\omega + 2m^2\beta\omega$ and

$$U_{1dn} = \begin{pmatrix} \rho & \mathrm{dn}(x;m) \\ -\mathrm{dn}(x;m) & -\rho \end{pmatrix}, \tag{38}$$

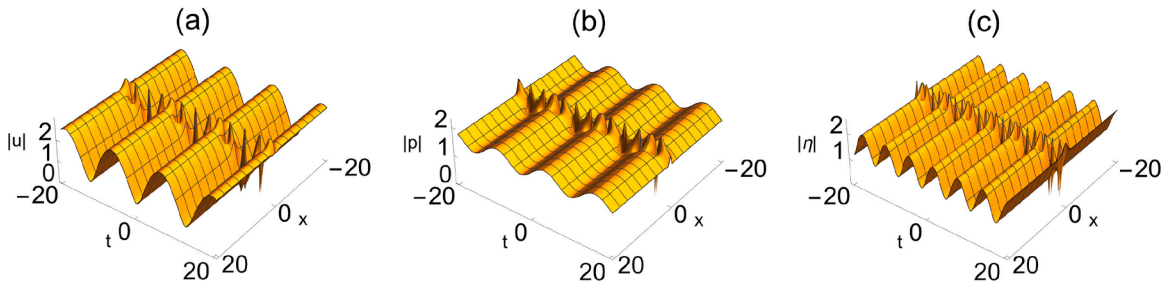


Fig. 1. 3D plots of one-breather on the periodic wave background in the Hirota and the Maxwell-Bloch equations with $q_0 = a_0 = 1, \rho_1 = 1, \alpha = 1, \beta = 0, \omega = 1$.

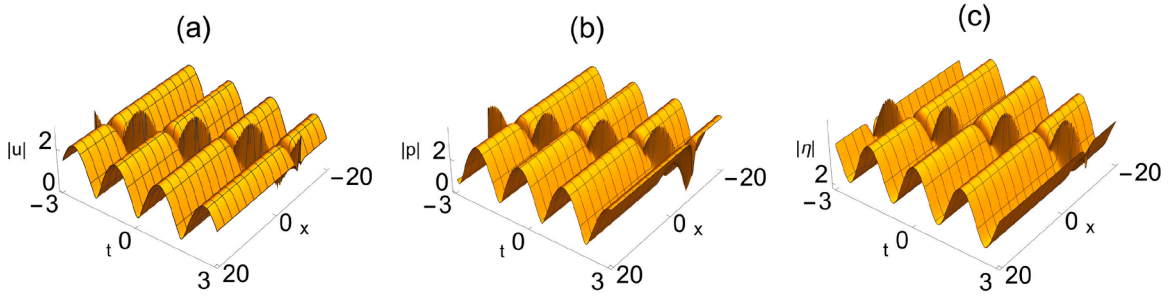


Fig. 2. 3D plots of one-breather on the periodic wave background in the Hirota and the Maxwell-Bloch equations with $q_0 = a_0 = 1, \rho_1 = 1, \alpha = 0, \beta = 1, \omega = 2$.

$$V_{1dn} = \begin{pmatrix} \tau'(-2+m^2+4\rho^2+2\mathrm{cn}^2(x;m)) & 4\rho\tau\mathrm{dn}(x;m) - 2m\tau\mathrm{cn}(x;m)\mathrm{sn}(x;m) - \iota' \\ -4\rho\tau\mathrm{dn}(x;m) - 2m\tau\mathrm{cn}(x;m)\mathrm{sn}(x;m) + \iota' & -\tau'(-2+m^2+4\rho^2+2\mathrm{cn}^2(x;m)) \end{pmatrix}, \tag{39}$$

where

$$\tau' = \frac{\alpha(\mathrm{i}\rho+\omega) - \beta(-2+m^2-4\rho^2+4\mathrm{i}\rho\omega)}{4(\rho-\mathrm{i}\omega)}, \quad \iota' = (m-1)m\beta\mathrm{dn}(x;m)\mathrm{sn}^2(x;m),$$

then we can get

$$\begin{pmatrix} \Psi_{1dn} \\ \Psi_{2dn} \end{pmatrix} = \begin{pmatrix} \frac{-2m\tau'\mathrm{cn}(x;m)\mathrm{sn}(x;m)+\mathrm{dn}(x;m)\sigma')}{\tau'(2m^2\mathrm{sn}^2(x;m)-\delta')} \Upsilon_{1dn}(x) e^{-\frac{\mathrm{i}}{2}\tau'\chi' t} + \Upsilon_{2dn}(x) e^{\frac{\mathrm{i}}{2}\tau'\chi' t} \\ \Upsilon_{1dn}(x) e^{-\frac{\mathrm{i}}{2}\tau'\chi' t} + \frac{2m\tau'\mathrm{cn}(x;m)\mathrm{sn}(x;m)+\mathrm{dn}(x;m)\sigma')}{\tau'(2m^2\mathrm{sn}^2(x;m)-\delta')} \Upsilon_{2dn}(x) e^{\frac{\mathrm{i}}{2}\tau'\chi' t} \end{pmatrix}, \tag{40}$$

where $\sigma' = 4\rho\tau' - (-1+m)m\beta\mathrm{sn}^2(x;m), \quad \delta' = m^2+4\rho^2+\frac{\mathrm{i}}{2}\chi', \quad \chi' = 2\sqrt{-m^4+16\rho^2-8m^2\rho^2-16\rho^4},$

$\Upsilon_{1dn}(x) = c_1(2m^2\mathrm{sn}^2(x;m)-\delta')^{\frac{1}{2}} e^{\rho(x+4m^2\delta'^{-1}-2)\Pi'+N'(x)},$

$\Upsilon_{2dn}(x) = c_2(2m^2\mathrm{sn}^2(x;m)-\delta')^{\frac{1}{2}} e^{-\rho(x+4m^2\delta'^{-1}-2)\Pi'+N'(x)},$

$\Pi'(s,\rho,m) = \int_0^s \frac{1}{(1-\rho\xi'^2)\sqrt{1-\xi'^2}\sqrt{1-\rho^2\xi'^2}}\mathrm{d}\xi', \quad N'(x) = \int \frac{(m-1)m\beta\mathrm{dn}^2(x;m)\mathrm{sn}^2(x;m)}{\tau'(2m^2\mathrm{sn}^2(x;m)-\delta')}\mathrm{d}x.$

5. Breathers on the periodic wave background

5.1. Breathers on the periodic wave background

Considering case 1, computing one-fold Darboux transformation, the two-fold Darboux transformation Eqs. (12) and (34) under the condition $c_1 = c_2 = 1, \lambda_1 = \mathrm{i}\rho_1$ and $\lambda_2 = \mathrm{i}\rho_2$ yields the following one-breather and two-breather on the periodic wave background.

Figs. 1–3 illustrate one-breather on the periodic background for the three potentials in Equation (1). Fig. 4 presents two-breather on the periodic background for Equation (1). It can be observed that by adjusting parameters α, β and ω, the breather frequency in Fig. 2 is significantly lower than that in Fig. 1. Among these periodic background configurations, the breathers and background waves in Figs. 1 and Fig. 2 both propagate along the t-axis. As the parameters α and β are adjusted, the breather in Fig. 3 propagates along the x-axis. Through observation of Fig. 4, the amplitudes of the two-breathers on the periodic background exhibit a substantial discrepancy. It should be noted that the parameter ρ_1 must be restricted to real values in this case.

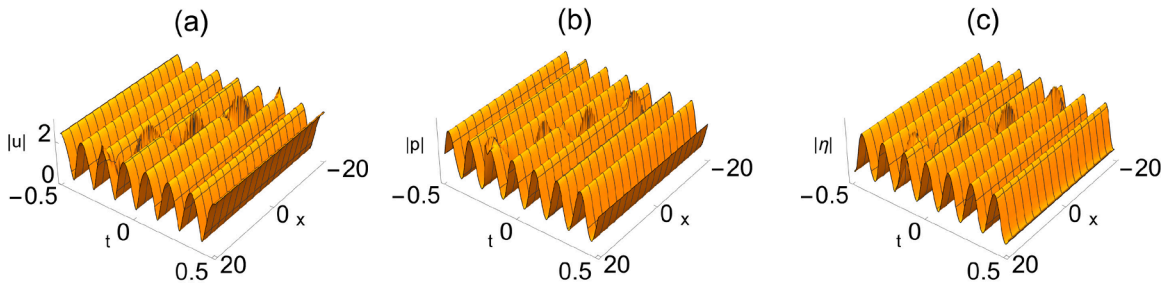


Fig. 3. 3D plots of one-breather on the periodic wave background in the Hirota and the Maxwell-Bloch equations with $q_0 = a_0 = 1, \rho_1 = 1, \alpha = 1, \beta = 30, \omega = 2$.

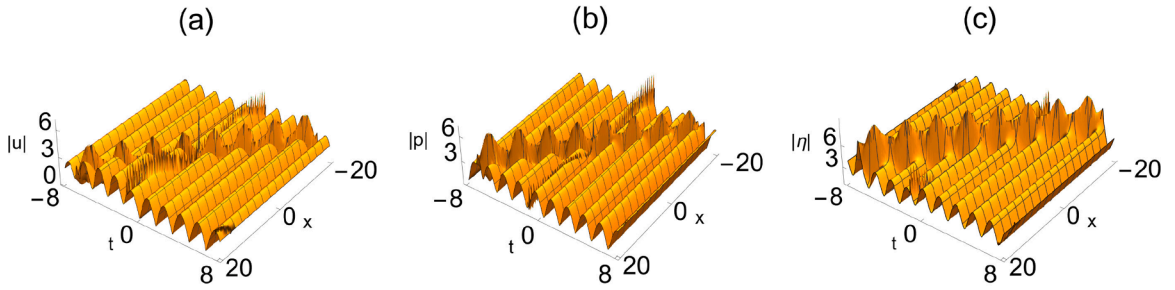


Fig. 4. 3D plots of two-breather on the periodic wave background in the Hirota and the Maxwell-Bloch equations with $q_0 = a_0 = 1, \rho_1 = 1, \rho_2 = 2, \alpha = 0, \beta = 1, \omega = 2$.

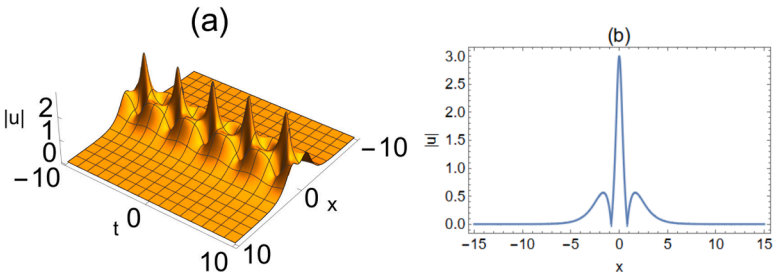


Fig. 5. (a):3D plots of the breathers on the soliton background in the Hirota and the Maxwell-Bloch equations with $\rho_1 = 1, m = 1, \alpha = 1, \beta = 0, \omega = 0$. (b):2D plots of the breathers on the soliton background in the Hirota and the Maxwell-Bloch equations with $\rho_1 = 1, m = 1, \alpha = 1, \beta = 0, \omega = 0, t = 0$.

5.2. *Breathers on the cn-periodic wave background*

Considering case 2, computing the one-fold Darboux transformation Eq. (12) and (37) under the condition $c_1 = c_2 = 1$ yields the following new nonlinear wave solutions

$$u[1]_{\mathrm{cn}} = e^{\mathrm{i}\kappa_1 t}\left(m\mathrm{cn}(x;m) + \frac{2\rho(A_1 + A_2)}{B_1 + B_2}\right), \tag{41}$$

where

$\Theta = \rho(x + (4m^2\delta^{-1} - 2)\Pi) + N(x)$,
$P(x) = \frac{1}{\tau}(m^2\beta - m^3\beta)\mathrm{sn}(x;m)\mathrm{cn}^2(x;m)$,
$W_1 = 2m^2\mathrm{sn}^2(x;m) - \delta$,
$W_2 = m^2\mathrm{sn}^4(x;m) - (1 + 4\rho^2\mathrm{sn}^2(x;m)) + 4\rho^2$,
$W_3 = -4(4\rho^2 + \delta - 1)m^2\mathrm{sn}^2(x;m) + 16m^2\rho^2 + \delta^2$,
$A_1 = 4mW_1(2\rho\mathrm{cn}(x;m)\cosh(2\Theta) - \mathrm{dn}(x;m)\mathrm{sn}(x;m)\sinh(2\Theta)) + P(x)^2e^{-\mathrm{i}\tau\chi t} + W_1^2e^{\mathrm{i}\tau\chi t}$,
$A_2 = 4m^2W_2e^{-\mathrm{i}\tau\chi t} + 2W_1P(x)\cosh(2\Theta) + 8m\rho P(x)\mathrm{cn}(x;m)e^{-\mathrm{i}\tau\chi t}$,
$B_1 = 8W_1m\rho\mathrm{cn}(x;m)\cos(\tau\chi t) + W_3\cosh(2\Theta) - 16\rho m^2\mathrm{cn}(x;m)\mathrm{dn}(x;m)\mathrm{sn}(x;m)\sinh(2\Theta)$,
$B_1 = 2W_1P(x)\cos(\tau\chi t) + 2mP(x)(4\rho\mathrm{cn}(x;m)\cosh(2\Theta) - 2\mathrm{dn}(x;m)\mathrm{sn}(x;m)\sinh(2\Theta)) + 2P(x)^2\cosh(2\Theta)$.

Furthermore, through the computation of Eqs. (16)–(17) and (41) with appropriately chosen parameter values, we obtain a variety of rich nonlinear wave solutions.

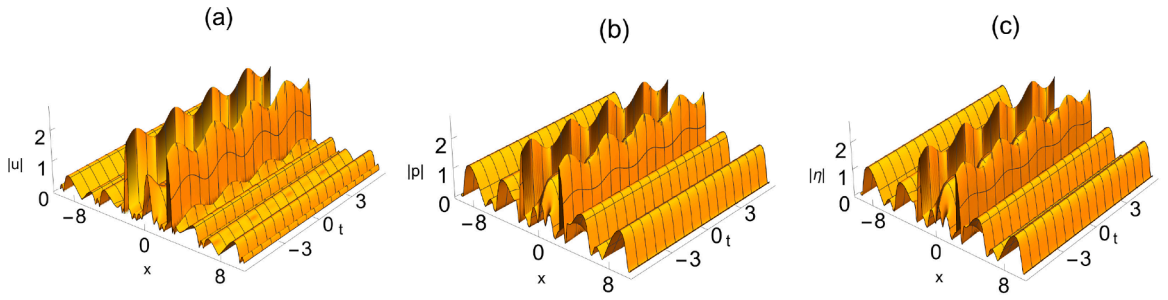


Fig. 6. 3D plots of breathers on the cn-periodic wave background in the Hirota and the Maxwell-Bloch equations with $\rho_1 = 1, m = \frac{\sqrt{2}}{2}, \alpha = 1, \beta = 0, \omega = 0$.

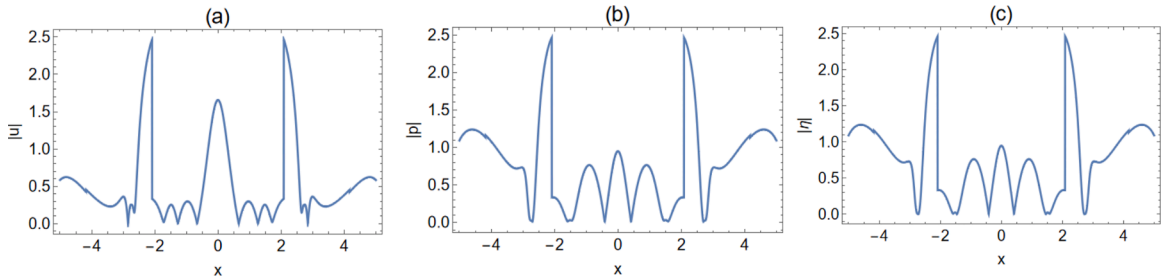


Fig. 7. 2D plots of breathers on the cn-periodic wave background in the Hirota and the Maxwell-Bloch equations with $\rho_1 = 1, m = \frac{\sqrt{2}}{2}, \alpha = 1, \beta = 0, \omega = 0, t = 0$.

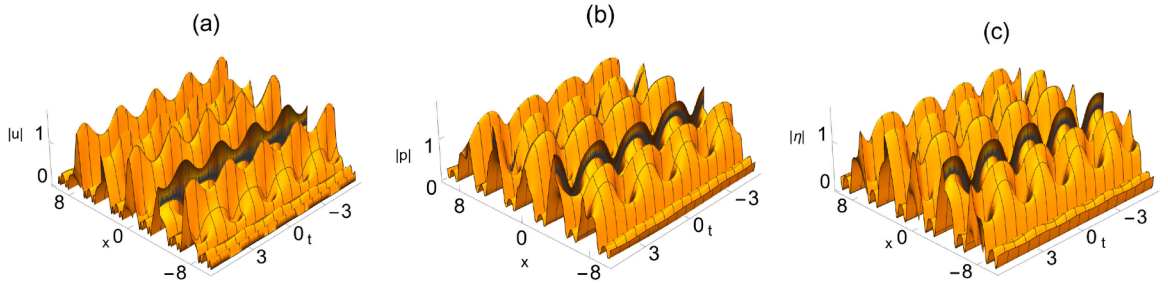


Fig. 8. 3D plots of breathers with periodic structure in the Hirota and the Maxwell-Bloch equations on the cn-periodic wave background with $\rho_1 = 1, m = \frac{1}{2}, \alpha = 1, \beta = 0, \omega = 0$.

Fig. 5 illustrates the breathers on the soliton background for Equation (1). Figs. 6 and Fig. 7 depict breathers on the cn-periodic background for the three potentials, while Figs. 8 and Fig. 9 show breathers with periodic structure on the cn-periodic background. It can be concluded from the figure that the breathers on the cn-periodic background appear as standing waves in the 3D plots, with their characteristic lines parallel to those of the background. When the parameter satisfies $\beta(-1 + 2m^2 + 4\rho^2) + \alpha\omega = 0$, breathers on the cn-periodic wave background can be obtained. As observed in Fig. 5, the breather effect is manifested on the soliton profiles. When the elliptic modulus $m = 1$, the periodic wave degenerates into a plane wave. A comparison between Figs. 6 and 8 reveals that the cn-periodic wave is irregular, and its background exerts an influence on the structure of breathers. Notably, an interesting phenomenon emerges as the parameter m decreases. The breathers initially existing on the cn-periodic background gradually evolve into those with a periodic structure. In other words, under the effect of the cn-periodic wave background, the breathers ultimately exhibit a periodic structure. In these cn-periodic backgrounds, the breathers propagate along the t-axis, while the background waves propagate along the x-axis. Moreover, the amplitude of the breathers is lower than that of the periodic wave in Fig. 7.

5.3. Breathers on the dn-periodic wave background

Considering case 3, computing the one-fold Darboux transformation Eq. (12) and (40) under the condition $c_1 = c_2 = 1$ yields the following new nonlinear wave solutions

$$u[1]_{\mathrm{dn}} = e^{\mathrm{i}\kappa_2 t}\left(\mathrm{dn}(x;m) + \frac{2\rho(A_1' + A_2')}{B_1' + B_2'}\right), \tag{42}$$

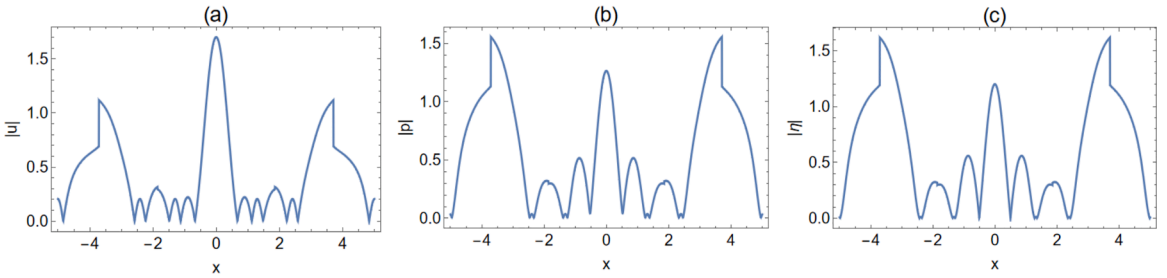


Fig. 9. 2D plots of breathers with periodic structure in the Hirota and the Maxwell-Bloch equations on the cn-periodic wave background with $\rho_1 = 1, m = \frac{1}{2}, \alpha = 1, \beta = 0, \omega = 0, t = 0$.

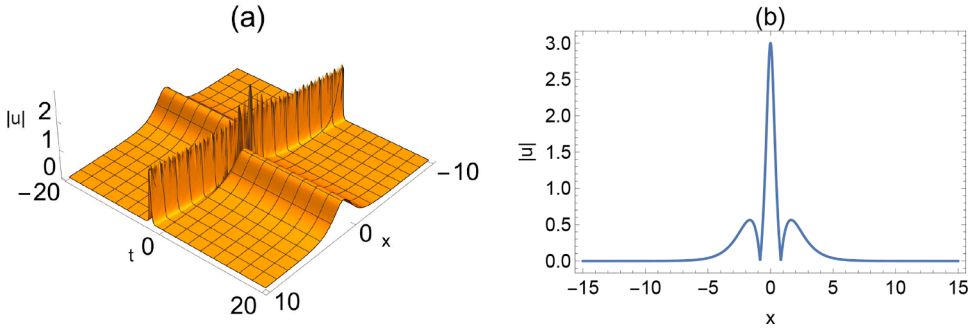


Fig. 10. (a):3D plots of two-soliton in the Hirota and the Maxwell-Bloch equations with $\rho_1 = 1, m = 1, \alpha = 1, \beta = 1, \omega = 0$. (b):2D plots of two-soliton in the Hirota and the Maxwell-Bloch equations with $\rho_1 = 1, m = 1, \alpha = 1, \beta = 1, \omega = 0, t = 0$.

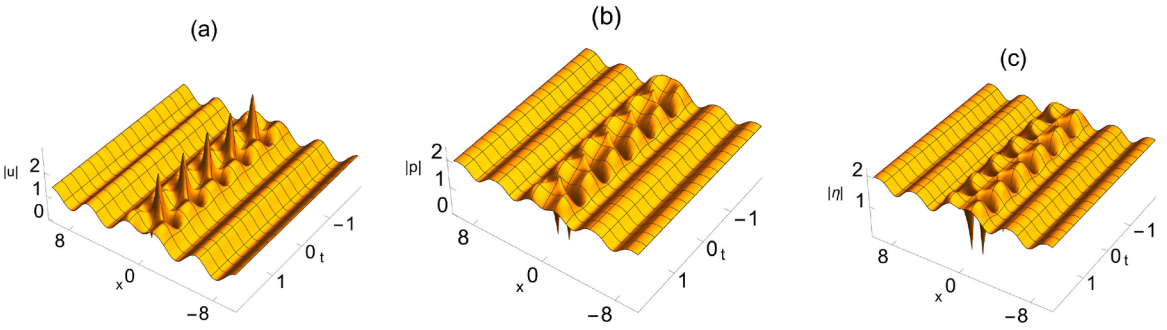


Fig. 11. 3D plots of breathers and anti-breathers on the dn-periodic wave background in the Hirota and the Maxwell-Bloch equations with $\rho_1 = 1, m = \frac{\sqrt{2}}{2}, \alpha = 8, \beta = 0, \omega = 0$.

where

$\Theta' = \rho(x + 4m^2\delta'^{-1} - 2)\Pi' + N'(x)$,

$P'(x) = -(m-1)k\beta \mathrm{dn}(x;m)\mathrm{sn}^2(x;m)$,

$W_1' = 2m^2\mathrm{sn}^2(x;m) - \delta'$,

$W_2' = m^2\mathrm{sn}^4(x;m) - (m^2 + 4\rho^2 m^2)$,

$W_3' = W_1'^2\tau'^2 + 4\tau'^2(4\rho^2\mathrm{dn}^2(x;m) + m^2\mathrm{cn}^2(x;m)\mathrm{sn}^2(x;m))$,

$A_1' = 4W_1'\tau'^2(2\rho\mathrm{dn}(x;m)\cosh(2\Theta') - m\mathrm{cn}(x;m)\mathrm{sn}(x;m)\sinh(2\Theta')) + P'(x)^2 e^{-i\tau'\chi' t} + W_1'^2\tau'^2 e^{i\tau'\chi' t}$,

$A_2' = 4\tau'^2 W_2' e^{-i\tau'\chi' t} + 2W_1' P'(x)\cosh(2\Theta') + 8\tau'\rho P'(x)\mathrm{dn}(x;m)e^{-i\tau'\chi' t}$,

$B_1' = 8W_1'\tau'^2\rho\mathrm{dn}(x;m)\cos(\tau'\chi' t) + W_3'\cosh(2\Theta') - 16\rho\tau'^2 m\mathrm{cn}(x;m)\mathrm{dn}(x;m)\mathrm{sn}(x;m)\sinh(2\Theta')$,

$B_2' = 2W_1' P(x)\cos(\tau'\chi' t) + 2\tau' P(x)(4\rho\mathrm{dn}(x;m)\cosh(2\Theta') - 2m\mathrm{cn}(x;m)\mathrm{sn}(x;m)\sinh(2\Theta')) + P'(x)^2\cosh(2\Theta')$.

Furthermore, through the computation of Eqs. (16)–(17) and (42) with appropriately chosen parameter values, we obtain a variety of rich nonlinear wave solutions.

Fig. 10 illustrates two-soliton of Equation (1). Figs. 11 and Fig. 13 depict breathers on the periodic background for the potential u, and anti-breathers on the periodic background for the potentials p and η, respectively. It can be concluded from the figure that

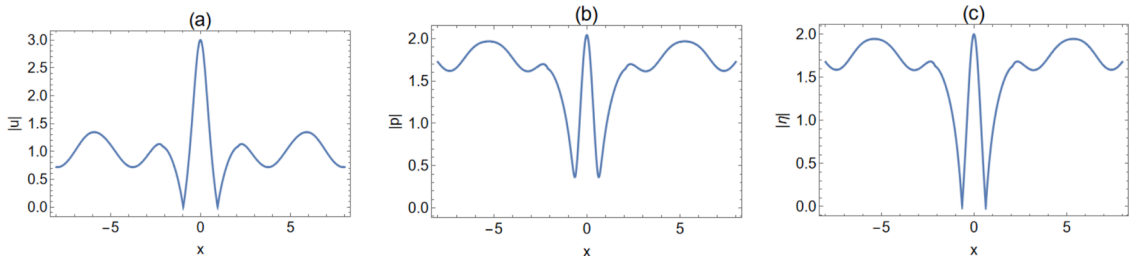


Fig. 12. 2D plots of breathers and anti-breathers on the dn-periodic wave background in the Hirota and the Maxwell-Bloch equations with $\rho_1 = 1, m = \frac{\sqrt{2}}{2}, \alpha = 8, \beta = 0, \omega = 0, t = 0$.

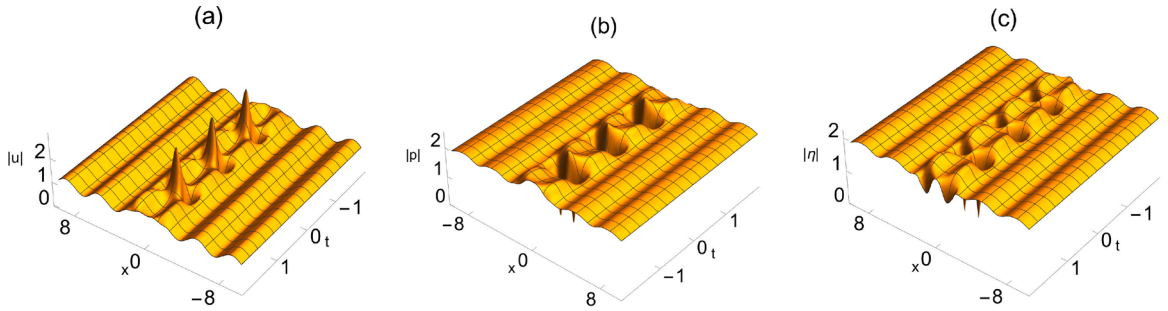


Fig. 13. 3D plots of breathers and anti-breathers on the dn-periodic wave background in the Hirota and the Maxwell-Bloch equations with $\rho_1 = 1, m = \frac{1}{2}, \alpha = 8, \beta = 0, \omega = 0$.

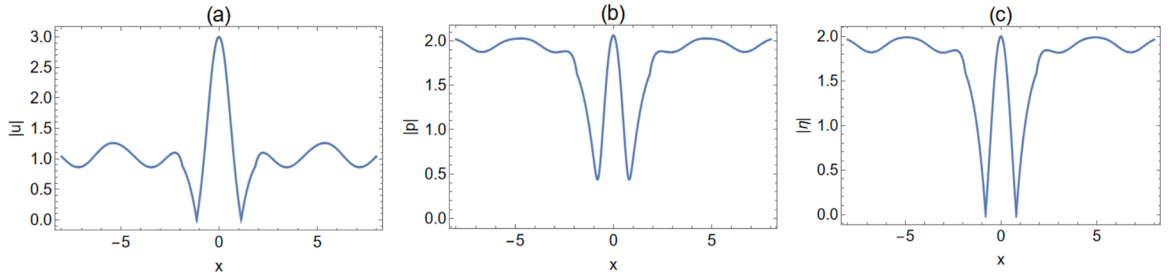


Fig. 14. 2D plots of breathers and anti-breathers on the dn-periodic wave background in the Hirota and the Maxwell-Bloch equations with $\rho_1 = 1, m = \frac{1}{2}, \alpha = 8, \beta = 0, \omega = 0, t = 0$.

the breathers on the dn-periodic background appear as standing waves in the 3D plots, with their characteristic lines parallel to those of the background. When the parameter satisfies $\beta(2 - m^2 + 4\rho^2) + \alpha\omega = 0$, breathers on the dn-periodic wave background can be obtained. As shown in Fig. 10, when the elliptic modulus $m = 1$, the periodic wave degenerates into a plane wave. A comparison of Figs. 6–9 and Figs. 11–14 reveals that, unlike cn-periodic waves, dn-periodic waves exhibit a regular pattern. In addition, as the elliptic modulus m decreases, breathers develop periodic structures under the influence of cn-periodic waves, whereas the dn-periodic wave background does not induce periodic structures in the breathers. Furthermore, the amplitudes of breathers on the cn-periodic background exceed those of the periodic background, while the amplitudes of breathers corresponding to potentials p and η on the dn-periodic background are smaller than those of the periodic background. Furthermore, Figs. 12 and 14 indicate that as m decreases, the amplitude of the periodic wave also decreases. A comparison among the potentials u, p and η reveals that the peaks of p and η are lower, while their troughs are wider. For breathers on dn-periodic backgrounds, the breathers propagate along the t-axis, whereas the background waves propagate along the x-axis.

6. Rogue waves on the periodic wave background

6.1. The eigenvalues of cn- and dn-periodic waves

We consider the following constraint

$$u = \varphi_1^2 + \varphi_2^{*2}, \tag{43}$$

where $F = (\varphi_1, \varphi_2)^{\mathrm{T}}$ is a nonzero solution of the Lax pair Eq. (2) with $\rho = \rho_1$ and φ_2^* is the complex conjugate of φ_2. Substituting Eq. (43) into (2) leads to the following finite-dimensional Hamiltonian system

$$\frac{d\varphi_1}{dx} = \frac{\partial H}{\partial \varphi_2}, \quad \frac{d\varphi_2}{dx} = -\frac{\partial H}{\partial \varphi_1}, \tag{44}$$

where

$$H = \rho_1 \varphi_1 \varphi_2 + \rho_1^* \varphi_1^* \varphi_2^* + \frac{1}{2}(\varphi_1^2 + \varphi_2^{*2})(\varphi_1^{*2} + \varphi_2^2). \tag{45}$$

Considering two of the conservation constants as

$$H_0 = \mathrm{i}(\varphi_1 \varphi_2 - \varphi_1^* \varphi_2^*), \tag{46}$$

$$H_1 = \rho_1 \varphi_1 \varphi_2 + \rho_1^* \varphi_1^* \varphi_2^* + \frac{1}{2}(|\varphi_1|^2 + |\varphi_2|^2)^2, \tag{47}$$

where H_0 and H_1 are constants associated with the variable x, and the relationship between them is $H = H_1 - \frac{1}{2}H_0^2$. Further, one has

$$\rho_1 \varphi_1^2 - \rho_1^* \varphi_2^{*2} = \frac{1}{2}u_x + \mathrm{i}uH_0. \tag{48}$$

Then, some other constants for u with $\rho_1 = \mathrm{ae} + i\phi$ can be summarized as

$$u^* u_x - u u_x^* = 2\mathrm{i}(2\phi - H_0)|u|^2 + 2\mathrm{i}H_0(H_0^2 + 2H_0\phi - 2H_1), \tag{49}$$

$$u_{xx} + 2|u|^2 u - 4(\mathrm{ae}^2 + \phi^2 + H_1 - \frac{1}{2}H_0^2 - 2\phi H_0)u = 2\mathrm{i}(2\phi - H_0)u_x, \tag{50}$$

$$\begin{aligned} |u_x|^2 \quad = \quad & -|u|^4 + 4(\mathrm{ae}^2 + \phi^2 + H_1 - \tfrac{1}{2}H_0^2 - 2\phi H_0)|u|^2 + 4\mathrm{ae}^2 H_0^2 \\ & -(H_0^2 + 2\phi H_0 - 2H_1)(5H_0^2 - 2\phi H_0 - 2H_1). \end{aligned} \tag{51}$$

With the help of Eq. (49) and (13), it is easy to know that the left-hand side of Eq. (49) becomes zero. Namely, we have

$$H_0 = 2\phi, \quad \phi(H_1 - 4\phi^2) = 0. \tag{52}$$

Substituting Eq. (13) into (50) and (51) and comparing the result with Eq. (14), we get

$$a_0 = 4(\mathrm{ae}^2 - 5\phi^2 + H_1), \quad b_0 = -4(H_1^2 - 12H_1\phi^2 + 32\phi^4 - 4\phi^2\mathrm{ae}^2), \tag{53}$$

through Eq. (52), we discuss the results in two cases: $\phi = 0$ or $\phi \neq 0$ and $H_1 = 4\phi^2$.

If $\phi = 0$, then relations Eq. (52) yield $H_0 = 0$, whereas relations Eq. (53) yield

$$a_0 = 4(\mathrm{ae}^2 + H_1), \quad b_0 = -4H_1^2. \tag{54}$$

As $b_0 < 0$, we choose dn in Eq. (22) and yield

$$H_1 = \pm\frac{1}{2}\sqrt{1 - m^2}, \quad \mathrm{ae} = \rho_1 = \rho_\pm = \frac{1}{2}(1 \pm \sqrt{1 - m^2}). \tag{55}$$

If $\phi \neq 0$, then relations Eq. (52) yield $H_0 = 2\phi$ and $H_1 = 4\phi^2$, whereas relations Eq. (53) yield

$$a_0 = 4(\mathrm{ae}^2 - \phi^2), \quad b_0 = 16\mathrm{ae}^2\phi^2. \tag{56}$$

As $b_0 > 0$, we choose cn in Eq. (21) and yield

$$\rho_1 = \pm\rho_\pm = \pm\frac{1}{2}(k \pm \mathrm{i}\sqrt{1 - m^2}). \tag{57}$$

6.2. Periodic and non-periodic solutions of the lax pair

Based on Eq. (43), (48) and $H_0 = 2\phi$, we get

$$\varphi_1^2 = \frac{2\rho_1 u + u_x}{2(\rho_1 + \rho_1^*)}, \quad \varphi_2^{*2} = \frac{2\rho_1^* u - u_x}{2(\rho_1 + \rho_1^*)}. \tag{58}$$

By using Eq. (13), we take

$$\varphi_1(x,t) = \Phi_1(x)e^{\frac{1}{2}\mathrm{i}\kappa t}, \quad \varphi_2(x,t) = \Phi_2(x)e^{-\frac{1}{2}\mathrm{i}\kappa t}, \tag{59}$$

substituting Eq. (13) and (60) into (59), we have

$$\Phi_1^2(x) = \frac{2\rho_1 Q + Q_x}{2(\rho_1 + \rho_1^*)}, \quad \Phi_2^{*2} = \frac{2\rho_1^* Q - Q_x}{2(\rho_1 + \rho_1^*)}. \tag{60}$$

Substituting Eq. (59) into (45) and (46), we get

$$\begin{aligned}&\rho_1\Phi_1\Phi_2+\rho_1^*\Phi_1^*\Phi_2^*-H+\tfrac{1}{2}Q^2=0,\quad \Phi_1^*\Phi_2^*=\Phi_1\Phi_2+\mathrm{i}H_0,\\&\Phi_1\Phi_2=-\tfrac{Q^2-2H+2\mathrm{i}H_0\rho_1^*}{2(\rho_1+\rho_1^*)}.\end{aligned}\tag{61}$$

From Eq. (60), we have

$$\Phi_1^2+\Phi_2^2=\frac{2\rho_1 Q}{\rho_1+\rho_1^*},\quad \Phi_1^2-\Phi_2^2=\frac{Q_x}{\rho_1+\rho_1^*}.\tag{62}$$

As for the dn-periodic solution $Q(x)=\mathrm{dn}(x;m)$ and taking based on $H_0=0$, $H=H_1=-\frac{1}{2}\sqrt{1-m^2}$, and taking $\rho_1=\rho_\pm=\frac{1}{2}(1\pm\sqrt{1-m^2})$, based on Eqs. (46)–(47), (61)–(62) are as follows

$$\Phi_1\Phi_2=-\frac{1}{4\mathrm{ae}}(Q^2+\sqrt{1-m^2}),\quad \Phi_1^2+\Phi_2^2=Q,\quad \Phi_1^2-\Phi_2^2=\frac{1}{2\mathrm{ae}}Q_x.\tag{63}$$

As for the cn-periodic solution $Q(x)=m\mathrm{cn}(x;m)$ and taking based on $H_0=2\phi$, $H_1=4\phi^2$, $H=2\phi^2$ and taking $\rho_1=\rho_\pm=\pm\frac{1}{2}(k\pm\mathrm{i}\sqrt{1-m^2})$, according to Eq. (46) and (47), (61) can be

$$\Phi_1\Phi_2=-\frac{1}{2m}(Q^2+\mathrm{i}m\sqrt{1-m^2}),\tag{64}$$

from Eq. (47) and (61), we get

$$\left(|\Phi_1|^2+|\Phi_2|^2\right)^2=1-m^2+|u|^2,\tag{65}$$

by use of $H_1=H_0^2$ from Eq. (57) and (64), we get

$$\left(|\Phi_1|^2-|\Phi_2|^2\right)^2=|u|^2-\frac{1}{m^2}|u|^2.\tag{66}$$

Taking the positive square root by Eq. (65) and the negative square root by Eq. (66), we know that

$$|\Phi_1|^2+|\Phi_2|^2=\mathrm{dn}(x;m),\quad |\Phi_1|^2-|\Phi_2|^2=-m\mathrm{sn}(x;m)\mathrm{cn}(x;m).\tag{67}$$

From Eq. (67), we have

$$|\Phi_1|^2=\frac{1}{2}(\mathrm{dn}(x;m)-m\mathrm{sn}(x;m)\mathrm{cn}(x;m)),\quad |\Phi_2|^2=\frac{1}{2}(\mathrm{dn}(x;m)+m\mathrm{sn}(x;m)\mathrm{cn}(x;m)),\tag{68}$$

according to Eq. (57) and (60), we get the following relationship

$$\Phi_1^2\Phi_2^{*2}=\frac{1}{4}[\mathrm{cn}^2(x;m)-\mathrm{sn}^2(x;m)\mathrm{dn}^2(x;m)+2\mathrm{i}\sqrt{1-m^2}\mathrm{sn}(x;m)\mathrm{cn}(x;m)\mathrm{dn}(x;m)],\tag{69}$$

which yields

$$\Phi_1\Phi_2^*=-\frac{1}{2}\mathrm{cn}(x;m)\mathrm{dn}(x;m)-\frac{\mathrm{i}}{2}\sqrt{1-m^2}\mathrm{sn}(x;m).\tag{70}$$

Suppose that $(\varphi_1,\varphi_2)^{\mathrm{T}}$ is the periodic solution of the Lax pair Eq. (2) with $\rho=\rho_1$, and that $(\psi_1,\psi_2)^{\mathrm{T}}$ is the second linearly independent solution of the Lax pair Eq. (2) with the same $\rho=\rho_1$, where ψ_1 and ψ_2 are non-periodic and have the following forms

$$\psi_1=\frac{\theta-1}{\varphi_2},\quad \psi_2=\frac{\theta+1}{\varphi_1},\tag{71}$$

where θ is a function of x, y and t to be determined.

Substituting Eq. (71) into Eq. (2) yields

$$\theta_x=\theta\frac{u\varphi_2^2-u^*\varphi_1^2}{\varphi_1\varphi_2}+\frac{u\varphi_2^2+u^*\varphi_1^2}{\varphi_1\varphi_2}.\tag{72}$$

With the help of Eqs. (13) and (59), we have

$$\theta_x=\theta Q\frac{\Phi_2^2-\Phi_1^2}{\Phi_1\Phi_2}+Q\frac{\Phi_2^2+\Phi_1^2}{\Phi_1\Phi_2}.\tag{73}$$

Substituting Eqs. (61) and (62) into (73), we get

$$\left(\frac{\theta}{Q^2-2H+2\mathrm{i}H_0\rho_1^*}\right)_x=\frac{-4\rho_1Q^2}{(Q^2-2H+2\mathrm{i}H_0\rho_1^*)^2}.\tag{74}$$

by integrating Eq. (74) with respect to x, we get

$$\theta=(Q^2-2H+2\mathrm{i}H_0\rho_1^*)\left[\int_0^x\frac{-4\rho_1Q^2(y)}{(Q^2(y)-2H+2\mathrm{i}H_0\rho_1^*)^2}\mathrm{d}y+\theta_0(t)\right],\tag{75}$$

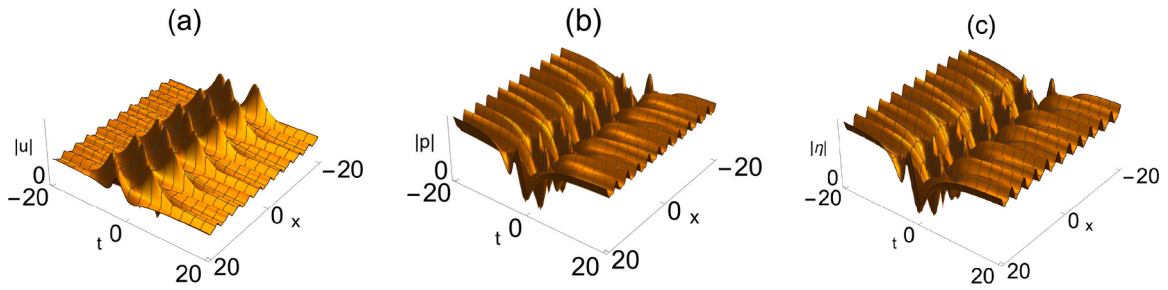


Fig. 15. 3D plots of the rogue waves on the cn-periodic wave background in the Hirota and the Maxwell-Bloch equations with $\rho_1 = 1, m = \frac{1}{2}, \alpha = 1, \beta = 0, \omega = 1$.

where $\theta_0(t)$ is an integral constant with respect to t.

Substituting Eqs. (71) into (3), a rather complex expression for θ_t will be derived

$$\theta_t = (Q^2(y) - 2H + 2\mathrm{i}H_0\rho_1^*)\lambda + \daleth, \tag{76}$$

where

$$\lambda = -\mathrm{i}\alpha - 4\beta\rho_1 - \frac{a_0\beta}{\rho_1 - \mathrm{i}\omega}, \quad \daleth = \frac{\mathrm{i}(2\kappa - a_0(\alpha - 4\beta\omega))\theta_x}{4(\rho_1 - \mathrm{i}\omega)}. \tag{77}$$

For the dn-periodic solution $Q(x) = \mathrm{dn}(x; m)$, under the condition of $a_0 = 2 - m^2$ and $b_0 = m^2 - 1$, we have $\daleth = 0$. For the cn-periodic solution $Q(x) = m\mathrm{cn}(x; m)$, under the condition of $a_0 = 2m^2 - 1$ and $b_0 = m^2 - m^4$, we have $\daleth = 0$. Thus, we rewrite Eq. (76) as

$$\theta_t = (Q^2(y) - 2H + 2\mathrm{i}H_0\rho_1^*)\lambda. \tag{78}$$

Comparing Eq. (77) with (78), we get

$$\theta_0(t) = \lambda t + \varrho, \tag{79}$$

where ϱ is the integral constant. Taking Eq. (78) into (74), we get the expression

$$\theta = (Q^2 - 2H + 2\mathrm{i}H_0\rho_1^*)\left[\int_0^x \frac{-4\rho_1 Q^2(y)}{(Q^2(y) - 2H + 2\mathrm{i}H_0\rho_1^*)^2}\mathrm{d}y + \lambda t + \varrho\right], \tag{80}$$

where λ is shown in Eq. (77).

6.3. Rogue waves on the cn-periodic wave background

According to the one-fold Darboux transformation Eq. (12), we apply the Jacobi elliptic function cn to obtain the rogue waves of the Hirota and the Maxwell-Bloch equations on the cn-periodic wave background. First, we take the seed solution with $u(x,t) = Q(x)e^{\kappa_1 t}$, and choose $\rho_1 = \rho_+$. On this basis, substituting $(\Phi_{1,1}, \Phi_{1,1})^{\mathrm{T}} = (\psi_1, \psi_2)^{\mathrm{T}}$ defined by Eq. (72) into (12), the form of the rogue waves of Eq. (1a) on the cn-periodic background is as follows

$$u_{\mathrm{rogue-cn}} = e^{\mathrm{i}\kappa_1 t}\left[m\mathrm{cn}(x;m) + \frac{2m(1 - 2\mathrm{i}\mathrm{Im}(\theta_{\mathrm{cn}}) - |\theta_{\mathrm{cn}}|^2)(\mathrm{cn}(x;m)\mathrm{dn}(x;m) + \mathrm{i}\sqrt{1-m^2}\mathrm{sn}(x;m))}{(1 + |\theta_{\mathrm{cn}}|^2)\mathrm{dn}(x;m) + 2m\mathrm{Re}(\theta_{\mathrm{cn}})\mathrm{sn}(x;m)\mathrm{cn}(x;m)}\right], \tag{81}$$

where

$$\theta_{\mathrm{cn}} = (m^2\mathrm{cn}^2(x;m) + \mathrm{i}m\sqrt{1-m^2})\left[-2(m + \mathrm{i}m\sqrt{1-m^2})\int_0^x \frac{m^2\mathrm{cn}^2(y)}{(m^2\mathrm{cn}^2(y) + \mathrm{i}m\sqrt{1-m^2})^2}\mathrm{d}y + \lambda t + \varrho\right], \tag{82}$$

λ is shown in Eq. (77), ϱ is the integral constant, κ_1 is given by Eq. (21). Furthermore, through the computation of Eqs. (16)–(17) and (81) with appropriately chosen parameter values, we obtain rogue waves on the cn-periodic wave background.

From Fig. 15 and Fig. 16, we can see that the rogue waves on the cn-periodic wave background of the Hirota and the Maxwell-Bloch equations are mainly linear rogue waves. Fig. 15 and Fig. 16 illustrate the rogue cn-periodic waves with $m = \frac{1}{2}$ and $m = \frac{9}{10}$, respectively. Notably, among the cases studied, only the rogue periodic waves corresponding to the potential u with $m = \frac{1}{2}$ and $m = \frac{9}{10}$ attain their maximum amplitude at the origin; for all other rogue periodic waves, the amplitude reaches its minimum at this point. It can be clearly seen from the figures that as the elliptic modulus m increases, the amplitude of the periodic background wave increases while the frequency decreases. That is to say, the periodic background wave amplitudes of the three potentials are regular.

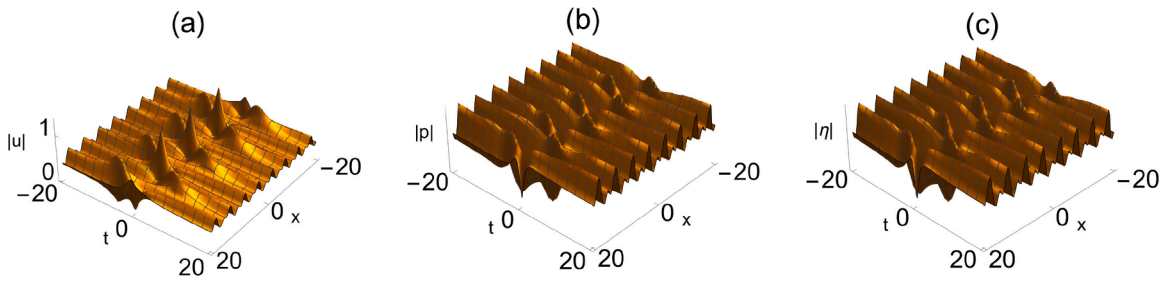


Fig. 16. 3D plots of the rogue waves on the cn-periodic wave background in the Hirota and the Maxwell-Bloch equations with $\rho_1 = 1, m = \frac{9}{10}, \alpha = 1, \beta = 0, \omega = 1$.

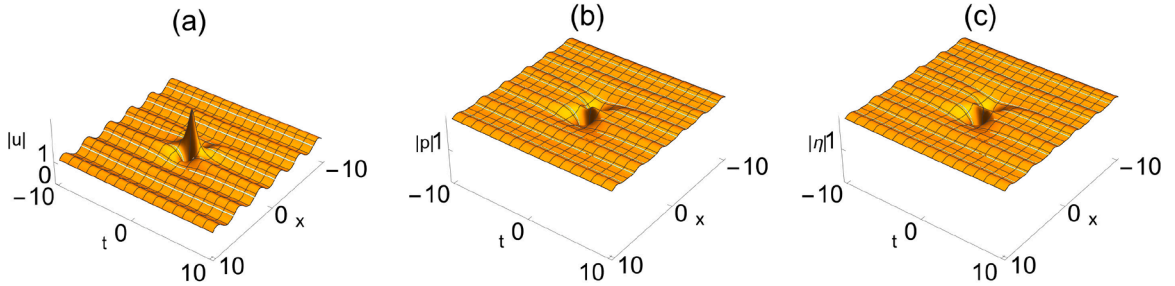


Fig. 17. 3D plots of the rogue waves on the dn-periodic wave background in the Hirota and the Maxwell-Bloch equations with $\rho_1 = 1, m = \frac{1}{2}, \alpha = 1, \beta = 0, \omega = 1$.

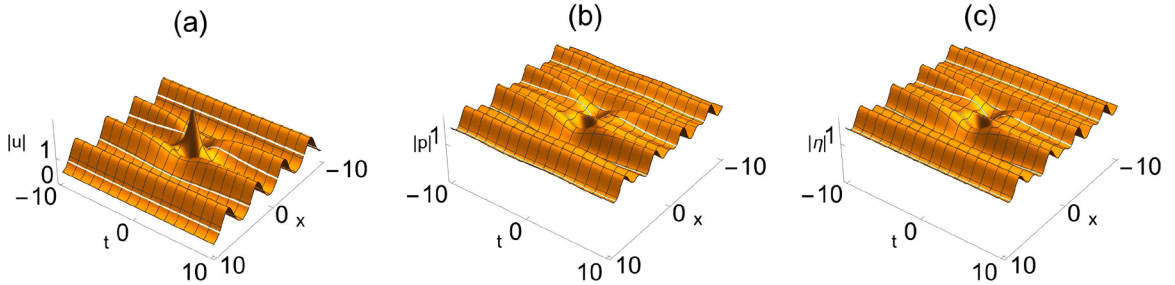


Fig. 18. 3D plots of the rogue waves on the dn-periodic wave background in the Hirota and the Maxwell-Bloch equations with $\rho_1 = 1, m = \frac{9}{10}, \alpha = 1, \beta = 0, \omega = 1$.

6.4. Rogue waves on the dn-periodic wave background

According to the one-fold Darboux transformation Eq. (12), we apply the Jacobi elliptic function dn to obtain the rogue waves of the Hirota and the Maxwell-Bloch equations on the dn-periodic wave background. First, we take the seed solution with $u(x,t) = Q(x)e^{\kappa_2 t}$, and choose $\rho_1 = \rho_+$, $H = -\frac{1}{2}\sqrt{1-m^2}$. On this basis, substituting $(\Phi_{1,1}, \Phi_{1,1})^{\mathrm{T}} = (\psi_1, \psi_2)^{\mathrm{T}}$ defined by Eq. (72) into (12), the form of the rogue waves of Eq. (1a) on the dn-periodic background is as follows

$$u_{\mathrm{rogue-dn}} = e^{\mathrm{i}\kappa_2 t}\left[\mathrm{dn}(x;m) + \frac{2(1-2\mathrm{i}\mathrm{Im}(\theta_{\mathrm{dn}}) - |\theta_{\mathrm{dn}}|^2)(\mathrm{dn}^2(x;m) + \sqrt{1-m^2})}{(1+|\theta_{\mathrm{dn}}|^2)\mathrm{dn}(x;m) + 2(1-\sqrt{1-m^2})\mathrm{Re}(\theta_{\mathrm{dn}})\mathrm{sn}(x;m)\mathrm{cn}(x;m)}\right], \tag{83}$$

where

$$\theta_{\mathrm{dn}} = (\mathrm{dn}^2(x;m) + \sqrt{1-m^2})\left[-2(1+\sqrt{1-m^2})\int_0^x \frac{\mathrm{dn}^2(y)}{(\mathrm{dn}^2(y)+\sqrt{1-m^2})^2}\mathrm{d}y + \lambda t + \varrho\right], \tag{84}$$

λ is shown in Eq. (77), ϱ is the integral constant, κ_2 is given by Eq. (22). Furthermore, through the computation of Eqs. (16)–(17) and (83) with appropriately chosen parameter values, we obtain rogue waves on the dn-periodic wave background.

From Fig. 17 and Fig. 18, it can be observed that, unlike the linear rogue waves on the cn-periodic background waves, rogue waves are obtained on the dn-periodic wave backgrounds of the Hirota and the Maxwell-Bloch equations. Fig. 17 and Fig. 18 illustrate the rogue waves on the dn-periodic waves with $m = \frac{1}{2}$ and $m = \frac{9}{10}$, respectively. Notably, among the cases studied, only the rogue periodic waves corresponding to the potential u with $m = \frac{1}{2}$ and $m = \frac{9}{10}$ attain their maximum amplitude at the origin; for all other rogue

Table 1
Classification of solutions for the Hirota and the Maxwell-Bloch equations.

Case	Solution Q	Parameters	Classification of solutions u
1	real number	$K=1, b_0=Q^4 4-a_0Q^2$	breather on periodic background
2	$K\mathrm{cn}(x;m)$	$K=m=1, b_0=m^2-m^4, a_0=2m^2-1$	breather on soliton background
3	$K\mathrm{cn}(x;m)$	$K=m, b_0=m^2-m^4, a_0=2m^2-1$	breather/rogue wave on cn background
4	$K\mathrm{dn}(x;m)$	$K=m=1, b_0=m^2-1, a_0=2-m^2$	two-soliton
5	$K\mathrm{dn}(x;m)$	$K=1, b_0=m^2-1, a_0=2-m^2$	breather/rogue wave on dn background

periodic waves, the amplitude reaches its minimum at this point. Additionally, the figures clearly demonstrate that as the elliptic modulus m increases, the amplitude of the periodic background wave increases, whereas its frequency decreases. In other words, similar to the cn-periodic background waves, the periodic background wave amplitudes of the three potentials exhibit a consistent regularity.

7. Conclusion

In the paper, we have investigated breathers on the soliton backgrounds, two-soliton, one-breather and two-breather on the periodic backgrounds by analyzing three special cases derived from a general solution of Equation (1). Compared to existing studies, breathers on the periodic backgrounds in multi-component systems remain relatively underexplored. For the cn-periodic case, a decrease in the elliptic modulus m leads to a transition from breathers on the cn-periodic background to breathers with periodic structure on the same background. When ρ is real, breathers and anti-breathers on the dn-periodic background can be obtained. Regarding propagation characteristics: for breathers on the periodic background, the background wave propagates along the t-axis, while the breather propagates along either the x-axis or the t-axis. In contrast, for breathers on cn- and dn-periodic backgrounds, the breather travels along the t-axis, while the background wave propagates along the x-axis. By combining the Darboux transformation with the nonlinearization of the Lax pair, rogue waves on the cn- and dn-periodic backgrounds of Jacobian elliptic functions are obtained. Specifically, the rogue waves on the cn-periodic background are linear, and the periodic background wave amplitudes of the three potentials exhibit a consistent regularity. Table 1 clearly illustrates the existence conditions of different types of solutions in the parameter space. These results contribute meaningful insights to the exploration of other integrable systems. It is hoped that the periodic wave solutions presented in this study will inspire new directions in future research.

CRediT authorship contribution statement

Yijie Zhao: Writing – original draft, Visualization, Software, Methodology, Data curation; **Zhaqilao:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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